

Section 4.1 control objective and front-position tracking draft

April 30, 2026

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4.1 Control objective and front-position tracking

The open-loop design problem was formulated as the selection of a cold-plate reference-temperature trajectory that promotes an approximately linear advance of the freezing front through the filled sample. For a candidate design vector θ , the workflow maps the requested reference trajectory to a predicted plate response, then to the simulated temperature field, and finally to the extracted freezing-front position,

$$\theta \rightarrow T_{\text{ref}}(t) \rightarrow T_{\text{plate}}(t) \rightarrow T(r, z, t) \rightarrow z_f(t) \rightarrow J(\theta).$$

Here, $T_{\text{ref}}(t)$ is the reference temperature supplied to the cryostage controller, $T_{\text{plate}}(t)$ is the plate-temperature response predicted by the calibrated cryostage model, $T(r, z, t)$ is the temperature field computed by the freezing solver, and $z_f(t)$ is the post-processed centreline freezing-front position. The trajectory parameterization used to construct $T_{\text{ref}}(t)$ from θ is described in Section 4.2.

During optimization, the inner cryostage response was represented by the calibrated first-order plate-temperature model,

$$T_{\text{plate},i} = \alpha_i T_{\text{plate},i-1} + (1 - \alpha_i)(gT_{\text{ref},i-1} + b),$$

$$\alpha_i = \exp\left(-\frac{\Delta t_i}{\tau}\right),$$

where τ , g , and b are obtained from cryostage characterization. This model represents the experimentally observed plate-temperature response to a requested reference trajectory, but it does not introduce feedback from the freezing front to the controller. The designed trajectory therefore remains an open-loop reference for the inner plate-temperature regulation layer. This model-based view is consistent with recent Stefan-system tracking formulations [Pet22] and with earlier inverse-solidification studies in which boundary cooling histories were designed to realize a desired interface motion [Kan95].

The objective was defined using front-position tracking rather than direct instantaneous front-velocity tracking. In the present workflow, the freezing front is obtained as a post-processed observable from the simulated field, whereas velocity would require numerical differentiation of the extracted trajectory. Because numerical differentiation of noisy sampled data is ill-posed and typically requires regularization, direct optimization of front velocity would be more sensitive to sampling, interpolation, thresholding, and discretization errors than tracking the front position itself [Bre20]. Front-velocity estimates were therefore treated as diagnostic quantities rather than primary optimization targets.

The target freezing behaviour was defined as a linear advance of the front through the filled sample height. This corresponds to an approximately constant average front velocity over the active interval without requiring velocity to be optimized directly. The reference front trajectory was written as

$$z_{f,\text{ref}}(t) = H_{\text{fill}} \text{clip}\left(\frac{t - t_0}{t_{\text{end}} - t_0}, 0, 1\right),$$

where H_{fill} is the filled sample height, t_0 is the start of the objective-evaluation window, and t_{end} is the design horizon. The corresponding implied average target velocity over the active interval is

$$v_{\text{ref}} = \frac{H_{\text{fill}}}{t_{\text{end}} - t_0}.$$

The use of t_0 allows the earliest part of the simulation to be excluded from the objective when needed, for example to avoid giving excessive weight to short initial transients immediately after filling.

For each candidate trajectory, the simulated front position was compared with the reference trajectory at the valid post-fill sample times. The main tracking term was defined as

$$J_{\text{track}}(\theta) = \frac{1}{N_t} \sum_{i=1}^{N_t} \left[\frac{z_f(t_i; \theta) - z_{f,\text{ref}}(t_i)}{H_{\text{fill}}} \right]^2,$$

where N_t is the number of time samples included in the objective calculation. Normalization by H_{fill} makes the tracking error dimensionless and expresses deviations relative to the sample height. This choice is consistent with standard model-based tracking formulations, in which quadratic penalties are applied to deviations between predicted outputs and desired reference trajectories [Sch21].

A smoothness regularization term was included to discourage abrupt changes in the requested reference-temperature trajectory. For the piecewise-linear parameterization used here, this term was computed from the slopes between consecutive trajectory knots,

$$J_{\text{smooth}}(\theta) = \frac{1}{N_k - 1} \sum_{j=1}^{N_k-1} \left[\frac{(T_{j+1} - T_j)/(t_{j+1} - t_j)}{\Delta T_{\text{ref}}/\Delta t_{\text{ref}}} \right]^2,$$

where T_j is the reference temperature at knot time t_j , N_k is the number of knots, Δt_{ref} is the full reference-trajectory duration, and ΔT_{ref} is the allowed reference-temperature range. This term acts as a regularizer on the requested thermal trajectory and is consistent with standard predictive-control practice, where penalties on input variation are used to avoid unnecessarily aggressive manipulated-variable motion [Sch21].

The scalar objective minimized during optimization was therefore

$$J(\theta) = w_{\text{track}} J_{\text{track}}(\theta) + w_{\text{smooth}} J_{\text{smooth}}(\theta) + w_{\text{comp}} J_{\text{comp}}(\theta),$$

where w_{track} , w_{smooth} , and w_{comp} are weighting factors. In the active linear full-process formulation, the tracking term provides the main optimization drive, while the smoothness term acts as a secondary regularizer on the requested plate-temperature trajectory. Completion-related penalties were retained in the implementation for alternative objective formulations, but they were not used as the main driver of the linear front-tracking objective unless stated otherwise in the optimization settings. Feasibility and admissibility constraints on the reference trajectory are described separately in Sections 4.2 and 4.3.

References

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